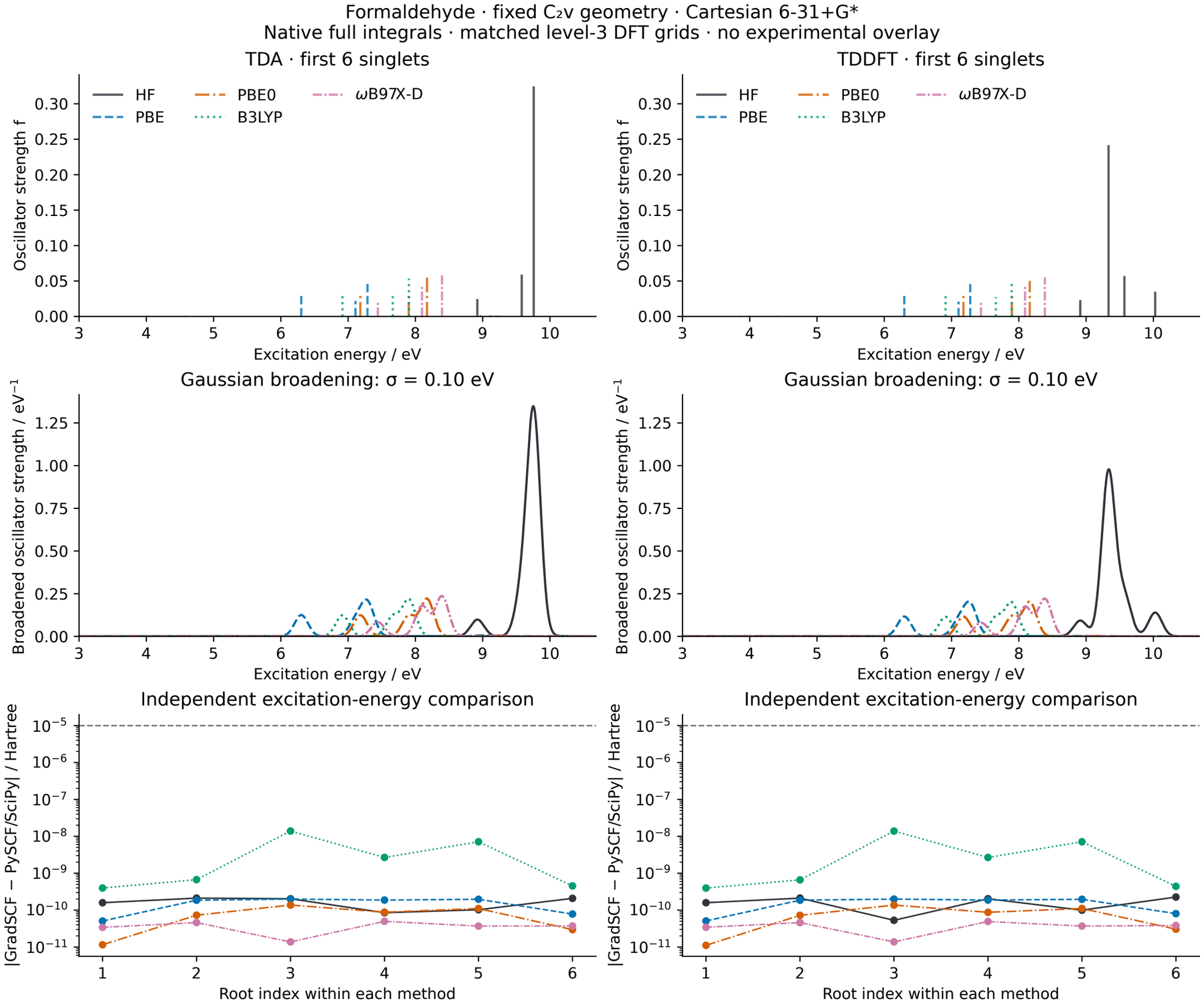
Formaldehyde · fixed C₂v geometry · Cartesian 6-31+G*
Native full integrals · matched level-3 DFT grids · no experimental overlay
TDA · first 6 singlets
TDDFT · first 6 singlets
HF
PBE
PBE0
B3LYP
ωB97X-D
Oscillator strength f
Excitation energy / eV
Gaussian broadening: σ = 0.10 eV
Broadened oscillator strength / eV⁻¹
Independent excitation-energy comparison
|GradSCF − PySCF/SciPy| / Hartree
Root index within each method